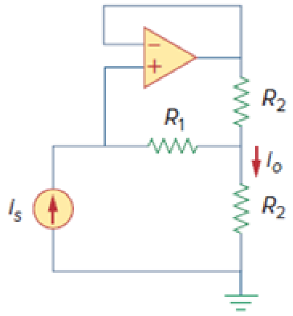
Chapter 5, Problem 91CP

Problem

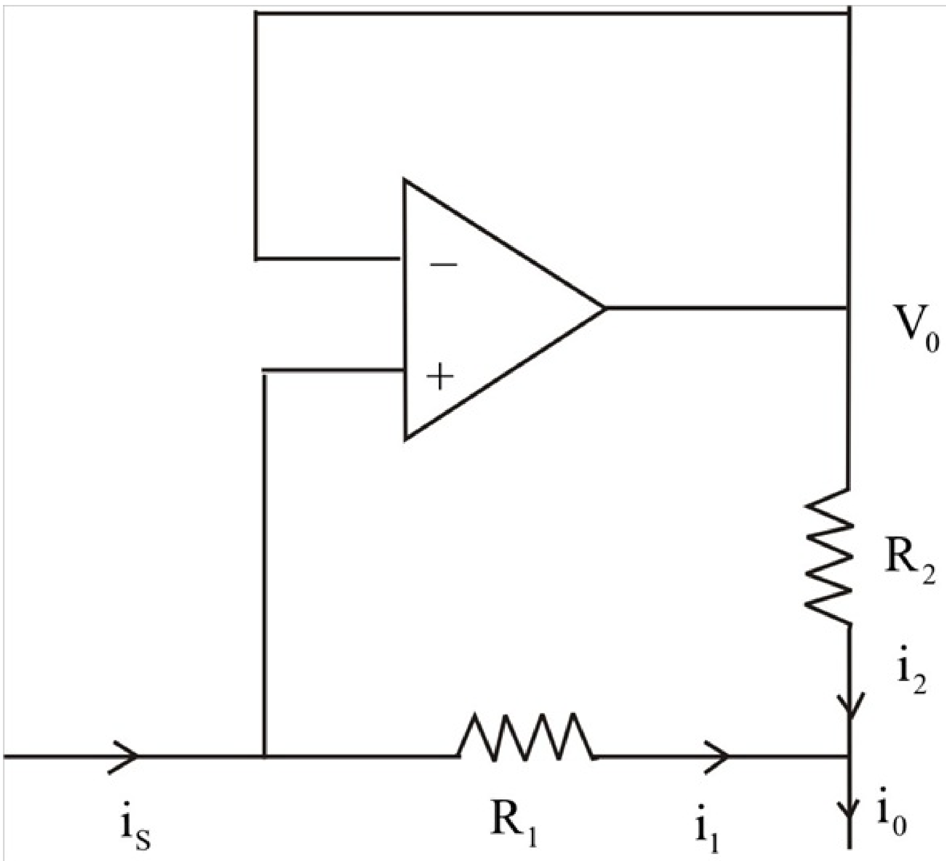
A noninverting current amplifier is portrayed in Fig. 5.108. Calculate the gain i_o/i_s. Take $R1 = 8$ kΩ and $R2 = 1$ kΩ.

Figure 5.108:



Step-by-step solution

Step 1 of 2



Step 2 of 2

$$i_0 = i_1 + i_2 \quad \rightarrow (1)$$

But $i_1 = i_S \quad \rightarrow (2)$

R1 and R2 have the same voltage, V0, across them,

$$R_1 i_1 = R_2 i_2 \;, \quad \textit{which leads to} \quad i_2 = \left(\frac{R_1}{R_2}\right) i_1 \quad \rightarrow (3)$$

Substituting (2) and (3) into (1) gives,

$$i_0 = i_S \left(1 + \frac{R_1}{R_2}\right) \; .$$

$$\frac{i_0}{i_S} = \left(1 + \frac{R_1}{R_2}\right) = 1 + \frac{8}{1} = 9 \; .$$